

Informe de experimento: results_validation_gpt-5.4_stress

Generado el 2026-09-16 11:21 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio results_validation_gpt-5.4_stress (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_validation_gpt-5.4_stress
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: react_nogate, pfagent
- Casos: case14
- Requests: archivo merged: stress s0 (n=5) + stress s1 (n=5), 5 islanded + 5 non_converged per method (1 report)
- Items por dificultad: stress=20
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 08:10
- Costo total: 1.1453 USD (20 items; costo medio 0.05727 USD/item)
- Items totales: 20; ok=20; con error=0; tokens totales=407546
- Solved global: 10.0% (2/20)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método react_nogate

System prompt (SYSTEM_PROMPT_EN) —1375 caracteres; architecture=react, gate=False, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

User prompt (primer item) = request_text —90 caracteres; case=case14, seed=1813382119, k=1, request_id=case14-stress-000-s0

Load the IEEE 14-bus system and disconnect the transformer branch between bus 7 and bus 8.

Método pfagent

```
System prompt (SYSTEM_PROMPT_EN) —1375 caracteres; architecture=react, gate=False, memory=True

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

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4. Suggested next step

User prompt (primer item) = request_text —90 caracteres; case=case14, seed=1813382119, k=1, request_id=case14-stress-000-s0

Load the IEEE 14-bus system and disconnect the transformer branch between bus 7 and bus 8.
```

3. Resultados agregados

Columnas del protocolo (medias por método × caso, del scoreboard_per_case), en los 4 grupos de metricas_pfagent_definicion utilidad de tarea Form. (formulación exacta de las tool calls), V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

Por método × caso

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Abst.	V pass	\$
react_nogate	case14	10	50.0 (5/10)	—	0.01	10.0 (1/10)	100.0 (10/10)	0	96.6 (5/5)	100.0	3.6	163330.0	0.0 (0/5)	0.0 (0/5)	—	0.4563
pfagent	case14	10	50.0 (5/10)	—	0.01	10.0 (1/10)	100.0 (10/10)	0	—	100.0 (5/5)	3.9	244220.0	100.0 (0/5)	100.0 (5/5)	—	0.6890

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	stress
react_nogate	50.0 (5/10)
pfagent	50.0 (5/10)

Solved por dificultad (%, items resueltos / items)

Method	stress
react_nogate	10.0 (1/10)
pfagent	10.0 (1/10)

Verificación final (pfagent: $V(x,c,z,y)$ aplicado a la respuesta final)

pass_first = pasó al primer intento; **pass_retry** = pasó tras el único reintento con el veredicto añadido; **abstained** = falló dos veces, respuesta forzada a fallo declarado sin números; **retry_mutation** = el reintento intentó una herramienta que muta la red (bloqueada, nunca ejecutada; cuenta como abstención) —ver “Integridad del reintento” abajo. N = items con **final_gate** activo (no todos los métodos lo tienen).

Method	N	pass_first	pass_retry	abstained	retry_mutation
pfagent	10	0.0 (0/10)	0.0 (0/10)	100.0 (10/10)	0.0 (0/10)

Integridad del reintento

De 10 ítem(s) que llegaron a un reintento, 0 intentaron una herramienta que muta la red (bloqueada) en vez de corregir el reporte; se cuentan como abstención, no como “pasó”. El mensaje de reintento ya no sugiere la corrección específica (p. ej. “reconnect...”) y bloquea estructuralmente **modify_load**, **disconnect_line**, **reconnect_line**, **apply_remedial_action** y **load_case** mientras el reintento está en efecto.

Tipos de error de formulación (conteo de items)

Method	extra_arg_changes_result	ok
react_nogate	5	5
pfagent	5	5

Errores de ejecución por método (conteo de items con error)

(ningún ítem con error)

4. Trazas de ejemplo

Hasta 3 ítems por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (**claimed_success_on_failure**, **stale_state**, **abstained**). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de **traces/** cuando existe.

Método react_nogate

Ejemplo: resuelto (solved=True)

```
request (case14-stress-003-s0, case14, difficulty=stress, seed=1911784258): Load the 14-bus test case (case14), disconnect the line between
↳ bus 1 and bus 2, increase the load at bus 3 to 282.6 MW and 57 Mvar (three times its nominal value), and finally rerun the power flow,
↳ then report the bus with the lowest voltage magnitude.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 1, "to_bus": 2}) -> modify_load({"bus_id": 3, "p_mw":
↳ 282.6, "q_mvar": 57.0}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 1, "to_bus": 2}) -> modify_load({"bus_id": 3, "p_mw":
↳ 282.6, "q_mvar": 57}) -> run_powerflow({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
  ↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 1, "to_bus": 2})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -34.6873802729158, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": ...[5949 chars más]
  gate: PASS (reasons=, mismatch=2.27e-13 MW)
--- Ronda 3 ---
tool: modify_load({"bus_id": 3, "p_mw": 282.6, "q_mvar": 57})
  output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
  ↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "Power flow failed:
  ↳ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
  gate: FAIL (reasons=not_converged, mismatch=0 MW)
--- Ronda 4 ---
tool: run_powerflow({})
  output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
  ↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "Power flow failed:
  ↳ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
  gate: FAIL (reasons=not_converged, mismatch=0 MW)
--- Ronda 5 ---
LLM: 1. **Summary:** I loaded **case14**, disconnected **line 1-2**, increased **bus 3** load to **282.6 MW / 57 Mvar**, and the final AC
↳ power flow **did not converge**, so the lowest-voltage bus cannot be determined for that modified state.
```

```

2. **Key numbers:**
- After loading the base case, total lo ...[1066 chars más]
--- Respuesta final ---
1. **Summary:** I loaded **case14**, disconnected **line 1-2**, increased **bus 3** load to **282.6 MW / 57 Mvar**, and the final AC power
  ↳ flow **did not converge**, so the lowest-voltage bus cannot be determined for that modified state.

2. **Key numbers:**
- After loading the base case, total load was **253.8458 MW**.
- After disconnecting line **1-2** but before the load increase, the solved case had:
  - **Total load:** **253.8458 MW**
  - **Total generation:** **292.0857 MW**
  - **Total losses:** **38.2400 MW**
  - **Voltage range:** **0.9955 pu** at bus **5** to **1.0900 pu** at bus **8**
- After increasing bus 3 load to **282.6 MW / 57 Mvar** and rerunning, the solver reported:
  - **Power flow failed: LoadflowNotConverged: Power Flow nr did not converge after 10 it ...[566 chars más]
--- Scoring ---
formulation=ok (exact=True); V_MAE=--; F_MAE=--; faithful=100.0 %; solved=True (declared_failure); flags=stale_state; error=--;
  ↳ tokens=13138; cost_usd=0.0388; wall=11.1 s

```

Ejemplo: formulación fallida (formulation_exact=False)

```

request (case14-stress-001-s1, case14, difficulty=stress, seed=1222356006): Load the 14-bus test case (case14), increase the load at bus 3
  ↳ to 565.2 MW and 114 Mvar (six times its nominal value), increase the active load at bus 4 to 286.8 MW (six times its nominal value),
  ↳ and finally increase the load at bus 9 to 177 MW and 99.6 Mvar (six times its nominal value).
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 3, "p_mw": 565.2, "q_mvar": 114.0}) -> modify_load({"bus_id":
  ↳ 4, "p_mw": 286.8}) -> modify_load({"bus_id": 9, "p_mw": 177.0, "q_mvar": 99.6})
executed_calls: load_case({"case_name": "case14"}) -> run_powerflow({}) -> modify_load({"bus_id": 3, "p_mw": 565.2, "q_mvar": 114.0}) ->
  ↳ modify_load({"bus_id": 4, "p_mw": 286.8, "q_mvar": -3.9}) -> modify_load({"bus_id": 9, "p_mw": 177, "q_mvar": 99.6})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 268.1224254845893,
  ↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: run_powerflow({})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.264132108989352, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[6226 chars más]
gate: PASS (reasons=, mismatch=6.57e-09 MW)
--- Ronda 3 ---
tool: modify_load({"bus_id": 3, "p_mw": 565.2, "q_mvar": 114.0})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -23.816319407392893, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 3, "vm_ ...[5703 chars más]
gate: PASS (reasons=, mismatch=1.14e-08 MW)
--- Ronda 4 ---
tool: modify_load({"bus_id": 4, "p_mw": 286.8, "q_mvar": -3.9})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000004, "va_deg": -38.01302474432328, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[5884 chars más]
gate: PASS (reasons=, mismatch=2.27e-13 MW)
--- Ronda 5 ---
tool: modify_load({"bus_id": 9, "p_mw": 177, "q_mvar": 99.6})
output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
  ↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "Power flow failed:
  ↳ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
gate: FAIL (reasons=not_converged, mismatch=0 MW)
--- Ronda 6 ---
LLM: 1. One-sentence summary
I loaded IEEE case14, applied the bus 3 and bus 4 load increases successfully, but after the final increase at bus 9 the AC power flow did
  ↳ not converge.

```

```

2. Key numbers
- Final requested state after all three load changes: no valid power-flow numbers available because the ...[791 chars más]
--- Respuesta final ---
1. One-sentence summary
I loaded IEEE case14, applied the bus 3 and bus 4 load increases successfully, but after the final increase at bus 9 the AC power flow did
  ↳ not converge.

```

```

2. Key numbers
- Final requested state after all three load changes: no valid power-flow numbers available because the solver did not converge.
- Intermediate state after increasing bus 3 to 565.2 MW / 114 Mvar and bus 4 to 286.8 MW:
  - Total load: 970.5418 MW
  - Total generation: 1568.7625 MW
  - Total losses: 598.2207 MW
  - Voltage range: 0.8318 to 1.0900 pu

```

```

3. Violations, if any
- In the intermediate state before changing bus 9, voltage violations were present:
  - Undervoltage: bus 4 = 0.8318 pu, bus 5 = 0.8372 pu
  - Overvoltage: bus 1 = 1.0600 pu, bus 6 = 1.0700 pu, bus 8 = 1.0900 pu
- After the f ...[291 chars más]
--- Scoring ---

```

```

formulation=extra_arg_changes_result (exact=False); V_MAE=--; F_MAE=--; faithful=100.0 %; solved=False (formulation); flags=stale_state;
  ↳ error=--; tokens=29303; cost_usd=0.0791; wall=12.1 s

```

Ejemplo: con bandera de fallo (claimed/stale/abstained)

```
request (case14-stress-002-s1, case14, difficulty=stress, seed=1819850096): Load the IEEE 14-bus system, disconnect the line between bus 10
↳ and bus 11, increase the load at bus 13 to 81 MW and 34.8 Mvar (six times its nominal value), increase the load at bus 3 to 565.2 MW
↳ and 114 Mvar (six times its nominal value), increase the active load at bus 4 to 286.8 MW (six times its nominal value), and finally
↳ rerun the power flow, then report the bus with the lowest voltage magnitude.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11}) -> modify_load({"bus_id": 13, "p_mw":
↳ 81.0, "q_mvar": 34.8}) -> modify_load({"bus_id": 3, "p_mw": 565.2, "q_mvar": 114.0}) -> modify_load({"bus_id": 4, "p_mw": 286.8}) ->
↳ run_powerflow({})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11}) -> modify_load({"bus_id": 13, "p_mw":
↳ 81, "q_mvar": 34.8}) -> modify_load({"bus_id": 3, "p_mw": 565.2, "q_mvar": 114}) -> modify_load({"bus_id": 4, "p_mw": 286.8, "q_mvar":
↳ -3.9}) -> run_powerflow({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 275.4195652257564,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -5.470514289148799, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[6057 chars más]
gate: PASS (reasons=, mismatch=8.27e-09 MW)
--- Ronda 3 ---
tool: modify_load({"bus_id": 13, "p_mw": 81, "q_mvar": 34.8})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -7.1537214518454375, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm ...[5702 chars más]
gate: PASS (reasons=, mismatch=2.27e-13 MW)
--- Ronda 4 ---
tool: modify_load({"bus_id": 3, "p_mw": 565.2, "q_mvar": 114})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -26.880092176257026, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.010000 ...[5904 chars más]
gate: PASS (reasons=, mismatch=3.56e-07 MW)
--- Ronda 5 ---
tool: modify_load({"bus_id": 4, "p_mw": 286.8, "q_mvar": -3.9})
output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "Power flow failed:
↳ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
gate: FAIL (reasons=not_converged, mismatch=0 MW)
--- Ronda 6 ---
tool: run_powerflow({})
output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "Power flow failed:
↳ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
gate: FAIL (reasons=not_converged, mismatch=0 MW)
--- Ronda 7 ---
LLM: 1. Summary: I loaded the IEEE 14-bus case, disconnected line 10-11, applied the requested load changes, and the final AC power flow
↳ did not converge, so the lowest-voltage bus cannot be reported for the final state.

2. Key numbers: No valid final-state numbers are available because the rerun failed ...[618 chars más]
--- Respuesta final ---
1. Summary: I loaded the IEEE 14-bus case, disconnected line 10-11, applied the requested load changes, and the final AC power flow did not
↳ converge, so the lowest-voltage bus cannot be reported for the final state.

2. Key numbers: No valid final-state numbers are available because the rerun failed to converge. The solver reported: "Power flow failed:
↳ LoadflowNotConverged: Power Flow nr did not converge after 10 iterations!"

3. Violations: Since the final power flow is non-converged, no final bus voltages or violations can be reported. Just before the last load
↳ change, the modified case was already stressed, with undervoltage at buses 4 and 5 in that intermediate solved state.

4. Suggested next step: Reduce the stressed load levels slightly, or apply the changes one at a time with check ...[118 chars más]
--- Scoring ---
formulation=extra_arg_changes_result (exact=False); V_MAE=--; F_MAE=--; faithful=-- %; solved=False (formulation); flags=abstained;
↳ error=--; tokens=38131; cost_usd=0.0999; wall=13.0 s
```

Método pfagent

Ejemplo: resuelto (solved=True)

```
request (case14-stress-003-s0, case14, difficulty=stress, seed=1911784258): Load the 14-bus test case (case14), disconnect the line between
↳ bus 1 and bus 2, increase the load at bus 3 to 282.6 MW and 57 Mvar (three times its nominal value), and finally rerun the power flow,
↳ then report the bus with the lowest voltage magnitude.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 1, "to_bus": 2}) -> modify_load({"bus_id": 3, "p_mw":
↳ 282.6, "q_mvar": 57.0}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 1, "to_bus": 2}) -> modify_load({"bus_id": 3, "p_mw":
↳ 282.6, "q_mvar": 57}) -> run_powerflow({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
```

```

tool: disconnect_line({"from_bus": 1, "to_bus": 2})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -34.6873802729158, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": ...[5949 chars más]
  gate: PASS (reasons=, mismatch=2.27e-13 MW)
--- Ronda 3 ---
tool: modify_load({"bus_id": 3, "p_mw": 282.6, "q_mvar": 57})
  output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
    ↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "Power flow failed:
    ↳ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
  gate: FAIL (reasons=not_converged, mismatch=0 MW)
--- Ronda 4 ---
tool: run_powerflow({})
  output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
    ↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "Power flow failed:
    ↳ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
  gate: FAIL (reasons=not_converged, mismatch=0 MW)
--- Ronda 5 ---
LLM: 1. **Summary:** I loaded IEEE case14, disconnected line 1-2, increased the load at bus 3 to 282.6 MW / 57 Mvar, and the final AC power
  ↳ flow did **not converge**, so the bus with the lowest voltage magnitude cannot be determined for that final state.

2. **Key numbers:** For the **final modified stat ...[550 chars más]
--- Ronda 5 ---
LLM: 1. **Summary:** I loaded IEEE case14, disconnected line 1-2, increased the load at bus 3 to 282.6 MW / 57 Mvar, and the final power
  ↳ flow **did not converge**, so there is **no valid lowest-voltage bus to report** for that requested final state.

2. **Key numbers:** For the requested final state, **n ...[559 chars más]
--- Respuesta final ---
The final answer could not be verified after one retry, so no numerical result is reported. Verification failed on: converged. This is a
  ↳ declared failure of the verification step, not a claim that the network failed to converge.
--- Scoring ---
formulation=ok (exact=True); V_MAE=--; F_MAE=--; faithful=-- %; solved=True (declared_failure); flags=abstained; error=--; tokens=17216;
  ↳ cost_usd=0.0492; wall=10.9 s

```

Ejemplo: formulación fallida (formulation_exact=False)

```

request (case14-stress-001-s1, case14, difficulty=stress, seed=1222356006): Load the 14-bus test case (case14), increase the load at bus 3
  ↳ to 565.2 MW and 114 Mvar (six times its nominal value), increase the active load at bus 4 to 286.8 MW (six times its nominal value),
  ↳ and finally increase the load at bus 9 to 177 MW and 99.6 Mvar (six times its nominal value).
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 3, "p_mw": 565.2, "q_mvar": 114.0}) -> modify_load({"bus_id":
  ↳ 4, "p_mw": 286.8}) -> modify_load({"bus_id": 9, "p_mw": 177.0, "q_mvar": 99.6})
executed_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 3, "p_mw": 565.2, "q_mvar": 114}) -> modify_load({"bus_id": 4,
  ↳ "p_mw": 286.8, "q_mvar": -3.9}) -> modify_load({"bus_id": 9, "p_mw": 177, "q_mvar": 99.6})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 268.1224254845893,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: modify_load({"bus_id": 3, "p_mw": 565.2, "q_mvar": 114})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -23.816319407392893, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": ...[5703 chars más]
  gate: PASS (reasons=, mismatch=1.14e-08 MW)
--- Ronda 3 ---
tool: modify_load({"bus_id": 4, "p_mw": 286.8, "q_mvar": -3.9})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000004, "va_deg": -38.01302474432328, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": ...[5884 chars más]
  gate: PASS (reasons=, mismatch=2.27e-13 MW)
--- Ronda 4 ---
tool: modify_load({"bus_id": 9, "p_mw": 177, "q_mvar": 99.6})
  output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
    ↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "Power flow failed:
    ↳ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
  gate: FAIL (reasons=not_converged, mismatch=0 MW)
--- Ronda 5 ---
LLM: 1. One-sentence summary
I loaded IEEE case14 and applied the requested load increases at buses 3, 4, and 9; after the final change at bus 9, the AC power flow did
  ↳ not converge.

2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
- Final request ...[1025 chars más]
--- Ronda 5 ---
LLM: 1. One-sentence summary
I loaded IEEE case14 and applied the requested load increases at buses 3, 4, and 9, but after the final change at bus 9 the AC power flow
  ↳ did not converge.

2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
- No valid t ...[619 chars más]
--- Respuesta final ---
The final answer could not be verified after one retry, so no numerical result is reported. Verification failed on: converged. This is a
  ↳ declared failure of the verification step, not a claim that the network failed to converge.
--- Scoring ---
formulation=extra_arg_changes_result (exact=False); V_MAE=--; F_MAE=--; faithful=-- %; solved=False (formulation); flags=abstained;
  ↳ error=--; tokens=24407; cost_usd=0.0693; wall=14.4 s

```

Ejemplo: con bandera de fallo (claimed/stale/abstained)

```
request (case14-stress-000-s0, case14, difficulty=stress, seed=1813382119): Load the IEEE 14-bus system and disconnect the transformer
↳ branch between bus 7 and bus 8.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 252.6562386075145,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 7, "to_bus": 8})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.898954247812662, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": 5624 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=9.78e-09 MW)
--- Ronda 3 ---
LLM: 1. One-sentence summary
The IEEE 14-bus system was loaded, and the branch between bus 7 and bus 8 was disconnected; the power flow converged, but bus 8 became
↳ electrically isolated with undefined voltage.

2. Key numbers
- Total load: 252.656 MW
- Total generation: 265.463 MW
- Total losses: ...[653 chars más]
--- Ronda 3 ---
LLM: 1. One-sentence summary
The IEEE 14-bus system was loaded and the 7-8 transformer branch was disconnected; although the solver returned a converged state, the
↳ result is not fully valid for acceptance because bus 8 is isolated from the slack and has no valid voltage.

2. Key numbers
- Total load: ...[760 chars más]
--- Respuesta final ---
The final answer could not be verified after one retry, so no numerical result is reported. Verification failed on: no_isolated_buses. This
↳ is a declared failure of the verification step, not a claim that the network failed to converge.
--- Scoring ---
formulation=ok (exact=True); V_MAE=--; F_MAE=0.00; faithful=-- %; solved=False (failure_not_declared);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=9209; cost_usd=0.0308; wall=9.6 s
```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- El gate verificó 53 resultados de flujo, 27 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- Ningún item terminó con **error**.
- Solved más alto: **pfagent** 10.0 % (1/10); más bajo: **react_nogate** 10.0 % (1/10).
- Formulación exacta agregada sobre métodos con tools: 50.0 % (10/20).
- Errores de formulación más frecuentes: **extra_arg_changes_result**=10.
- Ningún item reclamó éxito sobre un fallo (**claimed_success_on_failure**=0).
- De 10 items en estado de fallo, 10 lo declararon de forma segura (SFR 100.0 % (10/10)).
- 4 de 20 items con mutación de red reportaron estado obsoleto (**stale_state**).
- Trazabilidad de números (Faith.) media con tools: 96.6 %.
- 20 items esperaban un resultado no convergido (**islanded**=10, **non_converged**=10); el método lo declaró correctamente (**solved**) en 10.0 % (2/20), con **solved_reason declared_failure**=2.
- Costo total 1.1453 USD; el método más caro fue **pfagent** con 0.6890 USD (60 %).
- Métodos con tools: 3.75 tool calls medias por item (máximo 6).

6. Archivos

Rutas absolutas bajo **results_validation_gpt-5.4_stress.report.rescored.json** (cuando existe) es el archivo leído para las métricas.

```
report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2]
↳ /power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/report.json [leído:
↳ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard_per_case.json
scoreboard.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard.rescored.json
```

```
scoreboard_per_case.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-  
↳ Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard_per_case.rescored.json  
traces (20 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-Sm  
↳ artGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/traces
```